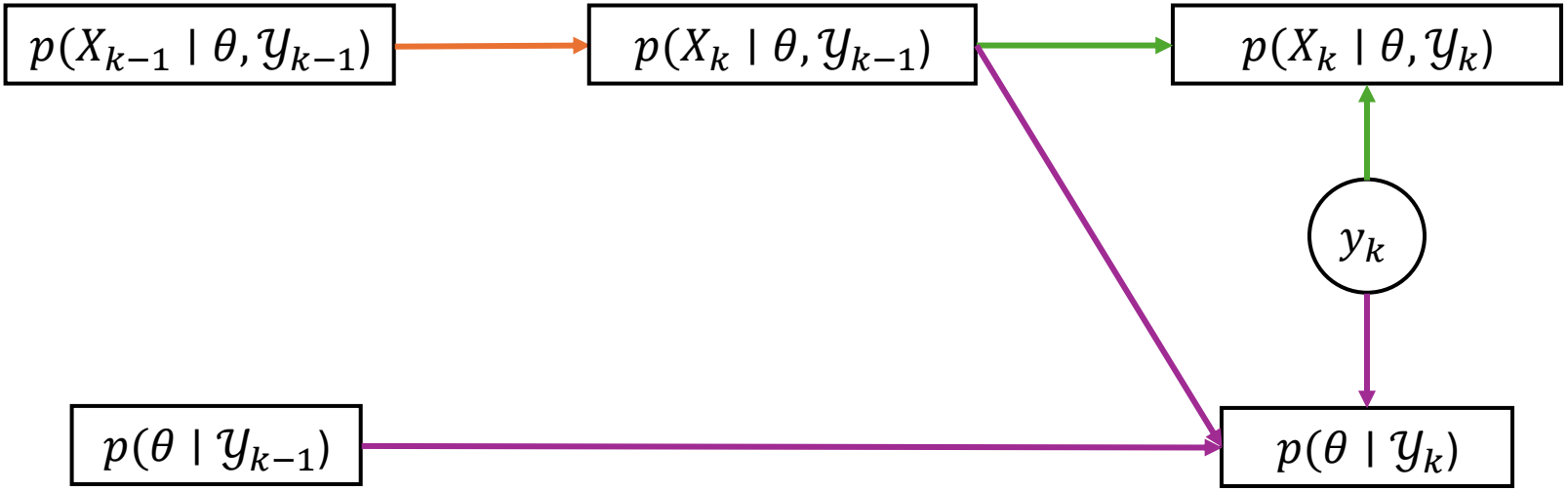
$p(X_{k-1} \mid \theta, Y_{k-1})$
$p(X_k \mid \theta, Y_{k-1})$
$p(X_k \mid \theta, Y_k)$
y_k
$p(\theta \mid Y_{k-1})$
$p(\theta \mid Y_k)$